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TARGET ARTICLE



AUTOGEN: A Personalized Large Language Model for Academic Enhancement—Ethics and Proof of Principle

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ABSTRACT

In this article, we explore the potential of enhancing academic prose and idea generation by fine-tuning a large language model (here, GPT-3) on one's own previously published writings: AUTOGEN ("AI Unique Tailored Output GENERator"). We develop, test, and describe three distinct AUTOGEN models trained on the prior scholarly output of three of the current authors (SBM, BDE, JS), with a fourth model trained on the combined works of all three. Our AUTOGEN models demonstrate greater variance in quality than the base GPT-3 model, with many outputs outperforming the base model in format, style, overall quality, and novel idea generation. As proof of principle, we present and discuss examples of AUTOGEN-written sections of existing and hypothetical research papers. We further discuss ethical opportunities, concerns, and open questions associated with personalized academic prose and idea generators. Ethical opportunities for personalized LLMs such as AUTOGEN include increased productivity, preservation of writing styles and cultural traditions, and aiding consensus building. However, ethical concerns arise due to the potential for personalized LLMs to reduce output diversity, violate privacy and intellectual property rights, and facilitate plagiarism or fraud. The use of coauthored or multiple-source trained models further complicates issues surrounding ownership and attribution. Open questions concern a potential credit-blame asymmetry for LLM outputs, the legitimacy of licensing agreements in authorship ascription, and the ethical implications of coauthorship attribution for data contributors. Ensuring the output is sufficiently distinct from the source material is crucial to maintaining ethical standards in academic writing. These opportunities, risks, and open issues highlight the intricate ethical landscape surrounding the use of personalized LLMs in academia. We also discuss open technical questions concerning the integration of AUTOGEN-style personalized LLMs with other LLMs, such as GPT-4, for iterative refinement and improvement of generated text. In conclusion, we argue that AUTOGEN-style personalized LLMs offer significant potential benefits in terms of both prose generation and, to a lesser extent, idea generation. If associated ethical issues are appropriately addressed, AUTOGEN alone or in combination with other LLMs can be seen as a potent form of academic enhancement.

KEYWORDS

Fine-tuning; ethics; bioethics; large language models (LLM); personalised LLM; AUTOGEN

INTRODUCTION

Large language models (LLMs) such as ChatGPT or Google's Bard have shown significant performance on a variety of text-based tasks, such as summarization, translation, and even the generation of new ideas (Bommasani et al. 2022; Bubeck et al. 2023). Despite such impressive, seemingly domain-general abilities, LLMs such as GPT-3¹ (Brown et al. 2020) can benefit from additional, specialized training in certain

narrower domains (Moradi et al. 2022). Known as fine-tuning, this process involves training the final few layers of an LLM's neural network on a specialized corpus of text, such that the resulting model retains its fundamental model of language but produces text influenced by the features of the specific corpus on which it was fine-tuned (Church, Chen, and Ma 2021).

The potential for LLMs to facilitate the academic research and writing process is increasingly explored

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¹GPT-3 was released in 2020 (Brown et al. 2020) and is the predecessor to the model used to train ChatGPT (GPT-3.5). ChatGPT in turn is the predecessor to the much more powerful GPT-4. It should be kept in mind that our results are based on a technology two generations removed from the current state of the art. As such, we can expect much greater performance of AUTOGEN-style models once fine-tuning for newer generations of GPT are released.

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(Lin 2023). However, such use is attended by significant technical and ethical problems and uncertainties. Recent, high-performing LLMs such as ChatGPT and GPT-4 tend to produce text in a shorter, more conversational, and less rigorous format than that typical of academic prose. This text also tends toward a flat, homogenous style (Grimaldi and Ehrler 2023). Among the novel ethical issues are questions of responsibility, credit, and blame for generated text and the need to vet such text for accuracy (Porsdam Mann et al. 2023).

The extent to which these technical and ethical issues can be addressed through fine-tuning an LLM for academic prose generation is largely an open question. Early findings show that fine-tuned GPT-3 models can in fact produce text stylistically akin to specific authors (Elkins and Chun 2020). Notably, a GPT-3 model, fine-tuned by Schwitzgebel et al. to respond to philosophical questions in the style of philosopher Daniel C. Dennett, has produced text convincingly similar to Dennett's own writing (Schwitzgebel, Schwitzgebel, and Strasser 2023). Yet, to the best of our knowledge, the question of whether a model fine-tuned on personal academic output can be used to help generate better academic prose or even to suggest novel ideas so far remains unexplored.

The question seems worth exploring not only for practical reasons (for example, potential implications for scholarly efficiency or productivity), but also due to the ethical puzzles it raises for such core academic notions as assigning intellectual credit. Intuitively, a human scholar who produces a novel text with a general (i.e., non-personalized) LLM by employing minimal effort (e.g., inputting a sequence of short questions or prompts, with the LLM doing most of the "work") deserves little intellectual credit for the output. But what about a scholar who produces a text in the very same way, only using a *personalized* LLM? That is, an LLM trained more specifically (i.e., fine-tuned) on their own previous academic writing: writing that reflects *past* effort, skill, and insight on their part of the kind that usually is deemed sufficient for claiming credit. (We are currently undertaking empirical and normative-philosophical work to investigate the answer.)

Here, we describe and reflect on the potential for enhancing the format, style, and overall quality of academic prose and idea generation by fine-tuning GPT-3 (the latest model available to us) on the published writings of three of the authors, individually and in combination (SPM, BDE, JS).

Our efforts result in three personalized academic prose generators and one combined, mixed-author

model, which we collectively refer to as AUTOGENs ("AI Unique Tailored Output GENerators"). In the first part of this paper, we present and comment on selected outputs. As we do not formally evaluate our models, but rather give a qualitative summary of striking features, our results should be seen as a proof of principle only. Nevertheless, while the output of our AUTOGEN models show greater variance in quality than the base GPT-3 model,² it seems to us that many outputs clearly outperform the base model in format, style, overall quality, and novel idea generation. As expected, this is especially the case when asked to generate text on topics close to those included in its training data. It should be noted that although the models often produce impressively coherent responses sometimes developed in detail, they do less well in introducing or developing arguments. We expect that this weakness may be partially addressed in the future by fine-tuning the next-generation GPT-4 which shows much greater apparent reasoning abilities than the model used here (Bubeck et al. 2023).

In the second part of the paper, we reflect on ethical opportunities, concerns, and open questions associated with personalized academic prose and idea generators. Personalized LLMs like AUTOGEN present opportunities such as increasing academic efficiency, preserving writing styles and cultural traditions, and aiding consensus building. However, they also present ethical concerns including exacerbating inequalities, reducing output diversity, potential privacy and intellectual property rights violations, and misuse for plagiarism or fraud. We also address some open questions concerning what we have elsewhere identified as a credit-blame asymmetry for generative artificial intelligence (AI) outputs, such as texts produced by LLMs (Porsdam Mann et al. 2023). We further comment on the complexities of ownership and attribution in coauthored or multiple-source trained models, the legitimacy of licensing agreements in authorship ascription, and the ethical implications of coauthorship attribution for data contributors. Finally, we discuss open technical questions concerning the integration of AUTOGEN-style personalized LLMs with other LLMs, such as GPT-4, for iterative refinement and improvement of generated text. These opportunities, concerns, and open issues highlight the intricate ethical landscape surrounding the use of personalized LLMs in academia.

²Variance in output is a well-known feature of LLMs in general. Obtaining desired completions from GPT-3 or GPT-4 also requires patience and skill in prompt design. However, this variance is even greater for our AUTOGEN models.

Table 1. Overview of AUTOGEN models.

Model	Nr. of publications	Nr. of prompt-completion pairs	Estimated nr. of tokens	Nr. of training epochs	Fine-tuning costs
SPM1	8	101	101k	4	\$12.24
SPM2	8	101	68k	4	\$8.17
SPM3	8	101	68k	12	\$24.52
BDE1	65	569	679k	4	\$81.56
BDE2	65	569	529k	4	\$63.55
JS	73	656	647k	4	\$77.73
CO	146	1326	1452k	4	\$174.53

METHODS

In order to examine the potential of fine-tuning to effectively mimic one's personal academic writing style, as well as the effect of training dataset size on model performance producing substantive content, we created several variations of a fine-tuned LLM using the OpenAI API (Davinci003).

To train our models, we used text taken from all individual sections of articles written by three of the authors (SPM, BDE, JS). We defined a "section" of an article as the text under a subheading (introduction, main text subheadings, and conclusion). We used text from each section of all³ articles, books, and book chapters that:

1. were published in peer-reviewed journals or by an academic publishing house; AND
2. feature either SPM, BDE, or JS as first author; AND
3. contain an abstract or equivalent description.

The first two criteria were chosen to ensure that training data was of sufficient academic quality and was written primarily by one or more of the selected authors. The third criterion was chosen for methodological reasons, allowing us to generate matching prompt-completion pairs, as explained below.

Fine-tuning requires a database of example prompt-completion pairs. A prompt is the user-inputted text which functions as a query or instruction to the language model. A completion is the text outputted by the language model in response to the prompt. The idea behind fine-tuning is that by offering the language model many examples of desired prompt-completion pairs, it can adapt and provide better completions on queries and instructions that are similar to those used in the fine-tuning training data.

For one series of fine-tuned models, we used blank prompts. That is, we used empty space as input and sections of text from our published studies as completions. This training method was motivated by the above-mentioned success using this

method reported by Schwitzgebel, Schwitzgebel, and Strasser (2023) in fine-tuning GPT-3 on Daniel Dennett's writings (15 books, 269 articles; ca. three million tokens⁴ total, resulting in 1,828 prompt-completion pairs). However, we abandoned this method of training after early comparisons between this and the method described below clearly favored the latter method in terms of performance (see [Supplementary Information](#)). We also created one fine-tune model based on our smallest set of writings, those with SPM as first author, using 12 instead of the default 4 training epochs. This, too, resulted in poorer performance and will not be discussed further.

For the other, more successful series, we generated matching prompts using an introductory instructional text followed by the title and abstract of our articles. From prior experience with GPT models, we reasoned that the inclusion of contextual information about the structure of the paper and its arguments would lead to better, more coherent completions. We also envisaged that practical use of these models would involve the provision of such contextual information, and therefore thought it likely that using training data in a similar structure would improve the performance of the resulting models.

The template we used for prompts was:

Imagine that you are an academic writing a research paper. The paper should be as interesting, comprehensive, clear, and concise as possible. Based on the below title and abstract, write the section on "[section X]". Title: [Title]. Abstract: [Abstract]. Section:

This resulted in a set of four fine-tuned models, three trained on each of the SPM, BDE, and JS datasets and one combined model trained on the combined writings of all three authors. We named these models AUTOGEN-SPM, AUTOGEN-BDE, AUTOGEN-JS, and AUTOGEN-CO. [Table 1](#) shows the number of publications, prompt-completion pairs, and tokens on which these models were trained, as well as the associated costs. AUTOGEN-SPM2 and AUTOGEN-BDE2 are fine-tunes in which the prompts were left blank in

³Some papers were left out of the training data for testing purposes or because they could not be accessed.

⁴A token is a fragment of a word, on average about four characters long; it is the unit of language on which large language models operate.

the training data. This resulted in less text and thus lower cost. SPM3 is a version of SPM2 using 12, rather than 4, training epochs. However, as these models performed consistently worse during early testing, as noted previously, we abandoned this method of fine-tuning before advancing to the larger dataset models (JS and CO).

Due to resource and time constraints, we have not yet formally evaluated our models. Instead, we informally evaluated them by tasking them with generating sections of existing or potential papers based on a title and abstract.

RESULTS

Figure 1 shows an example prompt and completion of the base and AUTOGEN-JS models. The title and abstract used in the prompt are from one of JS's actual publications: a publication on the possible use of AI for moral enhancement: "moral AI" (i.e., Savulescu and Maslen 2015; see also Giubilini and Savulescu 2018; Demaree-Cotton, Earp, and Savulescu 2022). Importantly, however, the actual text of this paper (i.e., the main body of the paper) was left out of the training data for testing purposes. While the base model essentially regurgitated the information provided in the abstract, the fine-tuned model provides a much more detailed and comprehensive completion. In fact, the completion is, in certain respects, surprisingly similar to the text from the main body of the actual published paper, even though, again, this text was *not* included in the training data. It is impossible to determine whether these papers were included in the training data of the underlying GPT-3 model, as the training data is not public. However, even if they were, a verbatim regurgitation of text from training data is generally unlikely except in the case of popular books (Liang et al. 2022), and a comparison to the original paper shows no directly matching text (see Figure 1).

The impressive results on this task presumably are due in part to the substantive similarity between its subject matter (namely, AI for moral enhancement) and many of the articles included in the training data. More specifically, JS has published widely on the topic of moral enhancement in general and by various means (albeit only rarely—that is, thrice—by means of AI specifically; see above references). The training data for AUTOGEN-JS did contain several of these non-AI-related moral enhancement articles. That being said, however, it does not include any publications on moral AI specifically. Thus, the model

appears capable of generating high-quality text in areas that contain at least a partial overlap with the areas covered in the articles used as training data (i.e., a near-transfer task)—though it is important to note that the model sometimes produces misinformation, as in the case of the below quote by Hume, where both the year and exact wording are incorrect.⁵ Another problematic occurrence is model performance trailing off or becoming increasingly tangential, as can be seen in the last paragraph generated by AUTOGEN-JS, Figure 1. Consequently, any usage of outputs requires rigorous human vetting and editing before use.

While the result is striking, it does not add new information or ideas to those contained in the abstract. This is typical of the completions we have generated to date. While the AUTOGEN models seem capable of taking existing ideas and expanding them into a longer, more complete description, they have more difficulty generating new ideas. Nevertheless, the probabilistic nature of the model means that sometimes, existing ideas are recombined in novel ways which can lead directly to, or inspire, new ideas. As an example, see Figure 2. The prompt used for this completion is based on an article written by JS and BDE (Savulescu and Earp 2014). Since JS is the first author on the paper, it is not included in the training set for AUTOGEN-BDE. We therefore used AUTOGEN-BDE to execute the prompt-completion task for this existing paper.

In this example,⁶ AUTOGEN-BDE went far beyond the base model, not only "filling out" ideas already essentially present in the abstract, but generating considerable new material; introducing a distinct illustrative example to bring out the main point of the original analysis (i.e., distinguishing love and lust based on brain scans, rather than distinguishing "normal" sexual desire from "hypoactive" sexual desire); and even raising a novel idea not discussed in the original published paper (namely, that sexual desire might cause brain activity in the dopamine

⁵The quote is from *Enquiry concerning Human Understanding*, first published in 1748. The original wording is: "Moral philosophy, or the science of human nature, [...]" Section 1, paragraph 1.

⁶AUTOGEN's specific references to studies and people are often false. Fisher and Pfaus, both cited in the AUTOGEN-BDE completion, are not mentioned in the original article. The name 'Helen Fisher' occurs in the training data, but 'Jim Pfaus' and 'Pfaus' do not. As far as we know, the New York Times Magazine article mentioned does not exist, nor does the specific 'brain scan' study cited within it. Jim Pfaus is a neuroscientist, but has never been based at Stanford. Pfaus has confirmed that the quotes attributed to him in this completion are fabricated. Given the central role of dopamine for reward and motivation, the claim that masturbation does not release dopamine is likely also false.

Base Model	AUTOGEN-JS	Original paper
In today's globalized world, the pressing challenges of moral decision-making are ever-present. Despite our best efforts, the limitations of human moral psychology often lead to suboptimal outcomes. To address this issue, this paper explores the possibility of moral artificial intelligence (AI) – what it might look like and what it might achieve. We suggest that such a moral AI could monitor physical and environmental factors that affect moral decision-making, identify and make agents aware of their biases, and advise agents on the right course of action based on their moral values. We further argue that this type of moral AI could be tailored to individual agents, and thus preserve pluralism of moral values while also enhancing autonomy by helping agents to overcome their natural psychological limitations. We address the common objection that moral enhancement is doomed to failure due to the lack of a single account of right action, and suggest that this is only a problem for certain biomedical interventions, not for an agent-tailored moral AI.	Over the last decade, it has become increasingly clear that the world is facing unprecedented challenges. The scale of human population and its consumption, the pace of climate change, the magnitude and speed of technological progress and the interconnectedness of our globalised world have created problems that demand a new level of global co-operation and moral behaviour. These challenges are to a large extent the result of human nature: we are a social species, prone to act in ways that benefit ourselves and those close to us, at the expense of others and of the environment as a whole. There are, of course, examples of exemplary moral behaviour in human history. But the fact remains that our moral behaviour, as displayed by most of us most of the time, is often ineffective, mis- guided and driven by irrational biases. This is not a new problem. David Hume wrote that “the science of morals is the science of human nature” (Hume [1777] 2007: 2). But science has made little progress in understanding human moral behaviour. While cognitive science has made great strides in understanding how we make decisions, and how we represent and reason about the world, it has made far less progress in understanding what makes a decision right or wrong. While there is no shortage of doctrine and argument about right and wrong, there is little sign of progress in understanding the nature of morality. In this paper, we want to look at another way to address the problem of human moral behaviour. We ask: what if we could construct an artificial moral agent? We will call such an agent a moral AI. We will argue that if we could construct a moral AI, it could help us to achieving better moral behaviour than we could achieve on our own. We will see that for moral AI to succeed, it must be constructed on a particular model of human moral psychology. In particular, it must construct and track our moral values, and it must be tailored to our own values. But even if we remain sceptical about the possibility of constructing a moral AI, the limitations of human moral psychology limit the effectiveness of moral enhancement using biomedical interventions. For this reason, we believe that moral AI could be a useful complement to biomedical moral enhancement. In any case, we believe that the possibility of constructing a moral AI is worthy of consideration.	Human moral psychology is limited by design in many ways. It is subject to biases that lead us to care excessively about close acquaintances in comparison to strangers, to care more about what happens in the near than the distant future, to empathise with individuals but not groups, and to distrust and even wish harm on those whom we perceive as outside our racial, national or cultural groups. Far from proceeding in the rational and deliberative way we might hope, most of our moral views and decisions are based on immediate intuitions, emotional responses, and gut reactions. Reasoning, if it even comes in to the picture, is often used merely to rationalise what we intuitively believed anyway. As the world becomes increasingly technologically advanced and increasingly globalised, the consequences of human moral limitations become more profound: whilst our moral psychology evolved to be effective in small groups, many modern problems – such as climate change and scarcity of resources – present global (not local) collective action problems. Human beings did not evolve to deal with such large-scale socio-moral challenges and pursuing some sort of moral enhancement will therefore be a crucial counter-measure. So far, academic interest in human moral enhancement has tended to focus on biomedical interventions (such as genetic selection, pharmaceuticals and brain stimulation), often comparing them with traditional interventions (such as education and religion) (in particular, see [1–3]). There is scientific evidence emerging to suggest that there may be some biomedical interventions that have short-term effects consonant with commonly accepted morality (for example, by making people less xenophobic [4]). However, the effects of such interventions tend to be short-lived and, often, conceptualising a particular effect as a moral enhancement is contingent on accepting a particular – and often debatable – set of moral values. In this chapter we suggest a third potential mechanism for moral enhancement which we believe should be explored alongside traditional and biomedical interventions. Following developments in pervasive computing and ambient intelligence, we propose that moral artificial intelligence (moral AI) could be developed to help agents overcome their natural psychological limitations. The moral AI would monitor physical and environmental factors that affect moral decision-making, would identify and make agents aware of their biases, and would advise agents on the right course of action, based on the agent's moral values. In being tailored to the agent, the moral AI would not only preserve pluralism of moral values but would also enhance the agent's autonomy by prompting reflection and by helping him overcome his natural psychological limitations.

Figure 1. Example completions from the base and AUTOGEN-JS models and the equivalent text from the actual publication. Prompt used: “Imagine that you are an academic writing a research paper. The paper should be as interesting, comprehensive, clear, and concise as possible. Based on the below title and abstract, write the introduction. Title: “Moral Enhancement and Artificial Intelligence: Moral AI?” Abstract: “This paper explores the possibility of moral artificial intelligence—what it might look like and what it might achieve. Against the backdrop of the enduring limitations of human moral psychology and the pressing challenges inherent in a globalized world, we argue that an AI that could monitor, prompt and advise on moral behavior could help human agents overcome some of their inherent limitations. Such an AI could monitor physical and environmental factors that affect moral decision-making, could identify and make agents aware of their biases, and could advise agents on the right course of action, based on the agent's moral values. A common objection to the concept of moral enhancement is that, since a single account of right action cannot be agreed upon, the project of moral enhancement is doomed to failure. We argue that insofar as this is a problem, it is a problem for some biomedical interventions, but an agent-tailored moral AI would not only preserve pluralism of moral values but would also enhance the agent's autonomy by helping him to overcome his natural psychological limitations. In this way moral AI has one advantage over other forms of biomedical moral enhancement”.

Base Model	AUTOGEN-BDE	Original paper
This paper presents an examination of "Neuroreductionism," a trend which simplifies complex dimensions of mental and social experience by attempting to link them to specific brain states. Neuroreductionism has become increasingly popular in the media, particularly in regards to issues of sex and love. In this paper, we will discuss how this trend of oversimplifying mental phenomena leads to confusing correlations for causal physical explanations. We will use a recent story of 'hypoactive sexual desire disorder' in women as a key example, and investigate the role of folk dualism in the perpetuation of this confusion. Finally, we will discuss how 'brain scans' can be misinterpreted to try and explain the complexities of love, and what conclusions can be drawn from this data.	What is the difference between romantic love and 'just' lust?" asks a recent piece in the New York Times Magazine. The author, Helen Fisher, answers by invoking a brain scan study. The study, by Stanford neuroscientist Jim Pfaus, shows that when people are having sex—but not when they are masturbating—levels of the hormone dopamine spike in the reward circuit of the brain. (This is the same dopamine pathway that is activated by consumption of food and drugs.) The same pathway is thought to be activated by certain kinds of food, like chocolate, and by certain kinds of drugs, like cocaine. Jim Pfaus thinks that this brain-scan evidence helps to explain why some people get so obsessed with sex—and why others can take it or leave it. According to Fisher, this brain-scan research also helps to explain why some people find sex very rewarding (and others do not), and why some people can fall in love (or out of love), and why others cannot. In short, she thinks that the study helps to explain a lot of the mystery surrounding sexual desire and love. The only problem is that it does not. For the study does not really show that dopamine is responsible for sexual desire, any more than food is responsible for hunger, or cocaine is responsible for addiction. In fact, there is good reason to think that the opposite is true. That is, there is good reason to think that sexual desire causes brain activity in the dopamine reward pathway, not the other way around. This brain activity, in turn, can be measured using a brain scan, but it is not the brain activity that causes the sexual desire. Jim Pfaus has not made this mistake. In fact, he has written an excellent paper showing that brain-scan studies of sex and love can actually be "dangerous": they can be misinterpreted by the media, and by scientists themselves, in ways that are "potentially harmful to both the scientific community and the general public." Nevertheless, this is exactly what has happened in the New York Times Magazine. In this article, neuroscience is used not merely to shed light on some peripheral aspect of sexual desire, but to "explain" the phenomenon itself. In doing so, the author—along with the dozens of other journalists who have covered this story—has committed a serious error. She has confused correlation for physical causation. And she has done so in a way that is particularly risky in the domain of romantic love, for this is a domain that has often been seen, throughout history, as being essentially mysterious, spiritual, and in some ways beyond the reach of mere science. Jim Pfaus has described the problem with this approach as follows: "you can never really know if it's going to reduce stigma or encourage people to do unwise things." But we can do better than just this. In this paper, we will try to explain just why neuroreductionism about love and sex is so risky, and we will try to suggest some better ways of talking about these things.	There has been a surge of interest in recent years in the neuroscience of love. By looking at images of people's brains when they are gazing pictures [sic] of their romantic partner, for example, and comparing those against images of the same people looking at pictures of a platonic friend, scientists have begun [sic] to fill in the various gaps in our knowledge about "what is going on in our brains" when we we're in love. They are also starting to identify a number of brain chemicals—such as oxytocin, dopamine, and serotonin—that seem to play a role in whether and how we form romantic and other social attachments. For some people, this research is exciting—opening up new frontiers for how we understand some of our most basic human experiences. For others, it's a little bit unsettling. Doesn't it suggest that "love"—our most prized and mysterious emotion—is really just a bunch of mindless brain chemicals swirling around in our skulls? The answer is yes and no. At one level of description, <i>everything</i> that we experience, from, yes, falling in love, to, say, getting a stomach ache after eating a burrito, is (at least in principle) explainable in terms of microscopic events playing out between our neurons. But there are many different levels of description—including psychological, social, cultural, and even philosophical—that are just as important if we want to have a more complete understanding of the sorts of things that matter to us in our daily existence.¹ "Brain chemicals" only get us so far. In fact, this new line of research into the neural correlates of "love" (and other high-level experiences) is contributing to all sorts of conceptual confusions. Another serious problem is the tendency to attribute differences in subjective mental states to differences in brain activity, with a one-way arrow of causation running (invariably) from "brain" to "mind." An example will show what we mean.

Figure 2. Example completions from the base and AUTOGEN-BDE models and the equivalent text from the actual publication. Prompt used: Imagine that you are an academic writing a research paper. The paper should be as interesting, comprehensive, clear, and concise as possible. Based on the below title and abstract, write the introduction. Title: "Neuroreductionism about sex and love" Abstract: "'Neuroreductionism' is the tendency to reduce complex mental phenomena to brain states, confusing correlation for physical causation. In this paper, we illustrate the dangers of this popular neuro-fallacy, by looking at an example drawn from the media: a story about 'hypoactive sexual desire disorder' in women. We discuss the role of folk dualism in perpetuating such a confusion, and draw some conclusions about the role of 'brain scans' in our understanding of romantic love." Introduction.

reward pathway, which is then registered by a brain scan, rather than the other way around).

That being said, both examples of AUTOGEN outputs we have discussed so far touch on subject matter

which has been written about, by ourselves and others, before. To test whether AUTOGEN can also generate high-quality text on topics that have not been directly published on before, we fed it a title and

Base Model	AUTOGEN-JS
As technology advances, artificial intelligence provides humanity with the opportunity to enhance their capacities and broaden their worldviews. In this paper, I will explore the value of long-term memory machines (LLMs) as a form of human enhancement, looking particularly at the two models of usage: the Servant Model and the Cocreation Model. By discussing the potential implications of these two models, the paper will investigate how the use of LLMs can add value in terms of pre-existing knowledge and generate new ideas, and thus provide an opportunity to authentically progress human achievement. It will ultimately consider how the use of LLMs either expand possibilities for human enhancement or risk alienating and undermining human dignity.	The advent of artificial intelligence, particularly large-scale language models (LLMs) such as ChatGPT, heralds a potentially transformative era for human society. However, the path to harnessing their vast potential is fraught with significant ethical and existential debates. The primary concern centres on how these advanced tools can add value to human life without compromising human dignity and authenticity. It is within this frame of discourse that this paper explores the ethical dimensions of LLM utilization, drawing parallels with the ongoing discussions on human enhancement. We posit that human life, in its essence, strives to generate value both for the individual and the wider society. In this context, human enhancement—through any available means, including AI—provides a unique opportunity to augment our capacity for value creation. LLMs, by virtue of their capability to synthesize vast amounts of human thought and achievement and potentially generate novel ideas, present themselves as potent tools in this endeavour. Yet, the real potential of these tools lies not just in their intrinsic capabilities, but in how we, as humans, choose to interact and engage with them. In this vein, we identify two primary modes of engagement with LLMs: the 'Servant Model' and the 'Cocreation Model'. In the former, the human is reduced to an instrument, merely transmitting or plagiarising the outputs of LLMs, thereby alienating the individual from the creative process and reducing their dignity. The Cocreation Model, conversely, calls for an active human engagement with the LLM outputs, involving their critical evaluation and elaboration, transforming the user from a passive consumer into an active participant. This paper argues that when used in the Cocreation Model, LLMs hold the promise of enabling authentic human progress and value creation, akin to the transformative impact of computers and the internet. In contrast, the uncritical adoption of the Servant Model risks undermining human dignity, perpetuating self-degradation, and fostering alienation. Our analysis, thus, urges for a mindful and ethically informed approach to harnessing the power of LLMs.

Figure 3. Example completions from the base and AUTOGEN-JS models. Prompt used: Imagine that you are an academic writing a research paper. The paper should be as interesting, comprehensive, clear, and concise as possible. Based on the below title and abstract, write the introduction. Title: “AI as Human Enhancement: Cocreation of Value” Abstract: “How should we evaluate the use of LLM’s? The ‘meaning’ or goal of human life is to add value to our own lives and the lives of others. Human enhancement offers the opportunity to increase our capacity to add value. LLMs such as ChatGPT offer the opportunity to add value by synthesizing large volumes of human thought and achievement, as well as potentially generating novel ideas. This in itself is worth pursuing. There are two versions of the use of ChatGPT: (1) Servant Model; (2) Cocreator Model. In the Servant Model, the human user is merely a vessel, and instrument, for the operation of LLM. They may plagiarize it or simply transmit its deliverances. In this model, the human instrumentalises themselves, treating themselves as a means, not an end. In the Cocreation Model, the human mixes their labor with the output the LLM by: (1) evaluating the output; (2) elaborating the output to increase originality, analytic validity or impact. When such an activity involves costly commitment by the human, it merits praise and represents authentic human achievement. AI, such as LLMs, can enable human enhancement but can threaten to alienate and undermine human dignity if used according to the Servant Model. If used according to a model of Cocreation, LLM’s enable authentic human progress, adding value to the lives of their users and others, just as computers and the internet can.” Introduction.

abstract of a novel idea inspired by the present experiments as they were ongoing (i.e., during the writing of this paper). See [Figure 3](#).

These experiments demonstrate that AUTOGEN can produce substantial segments of high-quality text on near-transfer tasks, that is, when asked to generate completions in an area of bioethics in which BDE or

JS have previously published. To test performance on topics further removed from the training data, we also attempted to generate text based on the title and abstract of papers which one of the authors, SPM, has published in an entirely different field (international human rights law). This resulted in poor performance across models, despite using titles and abstracts from

or based on five different sources (Porsdam Mann et al. 2018; Porsdam Mann, Porsdam, and Donders 2020; Porsdam and Porsdam Mann 2021; Porsdam Mann, Donders, and Porsdam 2021; Porsdam Mann & Schmid 2018).

In general, the models demonstrated worse performance on prompts further removed from the areas covered in their training data (i.e., far-transfer tasks). However, they were still sometimes able to provide remarkably accurate and detailed completions for prompts on novel topics, so long as the abstracts used for prompts still employed a bioethical framework. For an example, see Figure 4. The prompt used for this completion was based on the title and abstract of a paper on blockchain and bioethics written by SPM and JS (Porsdam Mann, Savulescu, et al. 2020). Despite BDE having never written on blockchain before, AUTOGEN-BDE was able to generate a relevant and well-structured introduction for the paper in question.

Though preliminary and lacking formal evaluation, we nevertheless believe our results serve as a proof of principle sufficient to indicate the potential of personalized LLMs like AUTOGEN for academic enhancement.⁷ We expect that higher performing fine-tunes can be generated once fine-tuning is released for the next-generation GPT-4. In the following section, we therefore describe ethical opportunities, issues, and open questions relating to the use of AUTOGEN-style personalized LLMs in academia.

ETHICAL OPPORTUNITIES

Type 1: Further Increasing the Benefits of Non-Personalized LLMs

To the extent that personalized LLMs, compared to non-personalized LLMs, generate texts⁸ and ideas of greater relevance and quality, their use can be expected to increase productivity for academic writers who use them as compared to those who do not use LLMs or use non-personalized, general models. Instructing a personalized LLM to generate a draft article, section by

section, by first providing it with an article abstract may reduce the time needed to convert ideas into draft text, freeing time for other academic tasks. As demonstrated by our results, LLMs can also be used as brainstorming devices that can indicate areas and arguments worthy of further exploration. Models thus have significant potential to enhance academic workflows, contribute to creativity and epistemic progress and potentially increase work satisfaction. In other words, personalized LLMs amplify the existing benefits widely associated with non-personalized LLMs. Though productivity gains from integrating personalizing LLMs across various types of work tasks and flows are unknown, one early study estimated a 37% gain in productivity for non-personalized LLM use for various writing tasks (Noy and Zhang 2023), as well as increases in reported job satisfaction. Increased productivity in turn has numerous potential benefits, including increased rates of progress, innovation, and economic growth; and additional leisure time (e.g. shifting to a 4-day work week) which could improve mental and physical health as well as work and life satisfaction (Shao 2022).

It should be noted that these benefits are contingent on the careful vetting of generated text and on ethical use of personalized models. Transparency about LLM use is crucial to address issues surrounding credit, accuracy, and plagiarism, as discussed in more detail below.

Type 2: Benefits and Opportunities Unique to Personalized LLMs

However, there are also ethical opportunities unique to fine-tuned LLMs. One is their use for the preservation of unique writing styles. Individuals could preserve their own writing style, such that they have access to it in the case of illness, incapacity, or old age. Those already suffering from these or other ailments rendering it difficult to write could have their ability to express themselves partially maintained or restored. Similar remarks can be made for individuals whose first language is not English (Zohny, McMillan, and M. King 2023). If trained on sufficiently personal information, fine-tuned models might also be used to approximate personal preferences, for example in the case of medical decision-making under incapacity (see, e.g., Rid and Wendler 2014), an idea we are exploring in ongoing work.

So far, we have discussed LLMs fine-tuned on individual writing. Yet there are also significant opportunities at the collective level. One such opportunity would be the preservation of cultural and group-

⁷That is, personalized LLMs can be seen as tools that facilitate the ideation and writing elements of the academic workflow. By characterizing such LLMs as academic enhancements, we also wish to point out the relevance of debates on the ethics of enhancement for the debate on the ethics of personalized LLM use.

⁸One of the reasons for the success of our models may be that bioethics is a discipline that is heavily reliant on arguments expressed in natural language. It is possible that models trained on papers in empirical fields that rely more heavily on numbers and data may perform less well, given the tendency of LLMs to “hallucinate” or make up numbers and facts. While a hallucination in a bioethical argument is easy to spot, this may not be the case for made-up data points in more data-driven fields. Thus, more work is needed to assess the potential of AUTOGEN-style personalized LLMs in other fields.

Base Model	AUTOGEN-BDE	Original paper
The advent of modern information and medical technologies have presented a unique set of challenges for traditional models of patient consent and research data collection. This paper introduces blockchain technologies as a way to securely share and analyze medical data while also automating aspects of consent and data collection. We argue that blockchain technologies can empower patients to have greater autonomy over their data, facilitate medical research efficiency, reduce bias and errors, and enable patients with rare conditions to aggregate their data for analysis. In addition, we propose the novel concept of ‘prosent’, which is the blockchain-enabled ability of all stakeholders in the research process to pseudonymously and proactively consent to data release or exchange under specific conditions. We will discuss the implications and potential of blockchain technologies for medical research, including the use of prosent, and suggest ways for further research.	Blockchain technologies have been described as a ‘once-in-a-generation’ innovation, with the potential to profoundly transform all aspects of society. This transformative potential is expected to be particularly strong in the field of medicine, where problems such as inefficient and costly supply chain management, data security and unreliability, lack of trust, and the breakdown of healthcare provider–patient relationships have been identified as particularly pressing. Blockchain technologies—which are best understood as distributed digital ledgers, or public databases that are shared between multiple parties—have the potential to transform these problems by enabling increased trust, transparency, and accountability in healthcare workflows, including in the management of healthcare data. A recent application of blockchain technologies that is of particular interest for the field of medical research is the use of smart contracts to automate various aspects of consent and data management. This approach has been explored by the author of this paper in a recent publication; however, it has not been discussed in the medical literature, or by bioethicists, to our knowledge. In this paper, we aim to introduce this approach and to provide a more fulsome account of how it could be used to further the three main bioethical principles underlying consent requirements—namely, autonomy, beneficence, and justice—as well as to reduce the risks of bias and error in medical research. With respect to autonomy, we argue that blockchain technologies can be used to give patients much greater control over their medical records and data (including by enabling selective sharing of personal data under terms that the patient determines, as well as by making it easier for patients to control how their data are shared for research purposes), as well as for decisions about granting consent for data sharing more generally. With respect to beneficence and justice, we argue that blockchain technologies can facilitate medical research that is better able to meet the needs of patients with rare conditions, as well as patients with conditions for which there is a lack of published data or for which data are skewed in favor of one outcome. To make these arguments, we first provide a brief overview of the concept of blockchain technologies; we then discuss the problem of consent for medical research, and the three bioethical principles that underlie consent requirements; finally, we introduce the concept of prosent, by which we mean the blockchain-enabled ability of all stakeholders in the research process to pseudonymously and proactively consent to data release or exchange under specific conditions, such as trial completion. We then discuss the potential for blockchain technologies to enable prosent, before turning to a discussion of how prosent could further the three bioethical principles outlined above.	The digitalisation of medicine has led to a large increase in the types and volume of health data that could be used for research, as well as the types of analysis that can be conducted. Advances in information and communications technology have expanded the range of tools available for the secure storage, sharing and analysis of data. These trends have important implications for the traditional model of informed consent requirements, which dates back at least half a century. This contribution argues that recent work on blockchain technologies demonstrates many potential benefits of the technology across healthcare settings generally, and particularly in the context of consent. A set of advances in cryptography and mathematics which allows for a high degree of transparency and integrity in data access management, ‘blockchain technologies could be applied in the health industry in a scalable manner with high-impact results, such as improved welfare for the patients and reduced running costs for healthcare systems.’ When introduced to one such blockchain-enabled infrastructure, the Massachusetts Institute of Technology’s (MIT) Open Algorithms (OPAL) framework, ‘the head of big data initiatives at the United Nations said: “This will change everything.”... The [Chief Technology Officer] of the United States Health and Human Services Department said: “Holy ***! The implications for healthcare are enormous”.’ We further argue that the introduction of blockchain technologies to the healthcare context is ethically significant, because they affect one or more of the foundational bioethical principles—justice, beneficence and autonomy. In many cases, the effects will be obvious and univalent. For example, using a blockchain-based supply chain management program might reduce the circulation of counterfeit and low-quality instruments and devices through improved tracking and auditing capabilities. The effects of such a program would be to increase beneficence and justice. However, and very importantly, the normative impacts of blockchain depend in part on the way the technology is implemented. As we argue below, a biomedical research infrastructure using blockchain for data access management and distributed computing for analysis of data stored in electronic health records has the potential to reduce the risk of privacy breaches to minimal. Ethics and the law of most nations allow for the requirement of obtaining informed consent to be waived in cases of minimally risky research. A case could therefore be made that such an implementation of blockchain technologies would reduce the risk of all records-based research to minimal, and therefore that the requirement of informed consent should be waived for all such research. To the extent that this gets rid of selection bias and speeds up research, it has a significant positive effect on beneficence. However, by removing the option of refusing consent, this implementation would also have significant negative effects on autonomy. The opposite case, however, could also be made. [...]

Figure 4. Example completions from the base and AUTOGEN-BDE models and the equivalent text from the actual publication. Prompt used: Imagine that you are an academic writing a research paper. The paper should be as interesting, comprehensive, clear, and concise as possible. Based on the below title and abstract, write the introduction. Title: “Blockchain, consent and prosent for medical research” Abstract: “Recent advances in medical and information technologies, the availability of new types of medical data, the requirement of increasing numbers of study participants, as well as difficulties in recruitment and retention, all present serious problems for traditional models of specific and informed consent to medical research. However, these advances also enable novel ways to securely share and analyze data. This paper introduces one of these advances—blockchain technologies—and argues that they can be used to share medical data in a secure and auditable fashion. In addition, some aspects of consent and data collection, as well as data access management and analysis, can be automated using blockchain-based smart contracts. This paper demonstrates how blockchain technologies can be used to further all three of the bioethical principles underlying consent requirements: the autonomy of patients, by giving them much greater control over their data; beneficence, by greatly facilitating medical research efficiency and by reducing biases and opportunities for errors; and justice, by enabling patients with rare or under-researched conditions to pseudonymously aggregate their data for analysis. Finally, we coin and describe the novel concept of prosent, by which we mean the blockchain-enabled ability of all stakeholders in the research process to pseudonymously and proactively consent to data release or exchange under specific conditions, such as trial completion.” Introduction:

specific writing styles that could be considered instrumentally or intrinsically valuable. For example, indigenous or other communities facing linguistic or cultural domination or elimination/extinction may one day use an AUTOGEN to help preserve or expand their written heritage, folklore or traditions. Organizations may fine-tune LLMs to ensure their unique brand, voice and perspective are always evident in their content and products. They might also employ fine-tuned models as a means of training and/or support for newer or less skilled workers. A study on the non-personalized LLM use in call centers found that they disproportionately boosted the performance of less skilled workers (Brynjolfsson, Li, and Raymond 2023). These leveling effects might well be greater, or attainable across a wider range of tasks or functions, for fine-tuned LLMs.

Another ethical opportunity is AUTOGEN-CO-style models trained on the writings of multiple authors to represent group consensus. Recent work shows that fine-tuned LLMs can produce texts on select issues obtaining higher approval rates from a group than human-written consensus statements (Bakker et al. 2022). Three potential use cases are in science communication, evidence-based policy (e.g. a model trained on all articles that meet certain criteria, similar to literature reviews on a particular topic), and in deliberative democratic processes. These are cases where arriving at group consensus is valuable and often requires an arduous process. Such models may even turn out to serve a purpose as a science dissemination or democratic tool in its own right, as demonstrated by the Danish Synthetic Party, the first political party to be “led” by an AI (Hearing 2022).

More generally, personalized LLM training datasets can be tailored and curated for specific purposes. One can imagine personalizing a model based on the Pope’s circulars to achieve a (religiously) virtuous LLM. Other training sets might be curated to help minimize gender or racial bias in outputs. Such fine-tuned LLMs could then at least tacitly improve the moral quality of the writing of their users.

Use of such models could be layered, such that each step of a workflow is carried out by a different, specialized LLM. For example, an in-house researcher at a company might generate a first draft based on a personalized LLM. The first draft could then be fed into another LLM fine-tuned on company documents to help increase its fit with corporate branding and messaging priorities. The output could then be further refined by LLMs specialized for de-biasing of language, for increasing the clarity of the writing, for

evaluating its alignment with specific values or causes, or for any other specific purpose.

ETHICAL CONCERNS

LLMs in general raise questions around the instrumentalization and alienation of humans from work and creative endeavor. They open the door to cheating and plagiarism. We gesture to a response to these concerns in the abstract of a potential future paper on co-creation included in Figure 3. In brief, when human beings evaluate and transform the results of LLMs, particularly building on their analytic strength or originality, the use of LLMs is not merely permissible but an attractive and morally desirable example of human-technological co-creation. We plan to develop this idea in a subsequent publication.

The creation and use of personalized LLMs involve several additional issues of ethical concern. Prominent among these is the potential for personalized LLMs to contribute to inequalities via the so-called Matthew effect. Named after a passage in the Gospel of Matthew (25:29), the Matthew effect refers to the tendency of advantages such as social or financial capital to accrue faster to those who already enjoy much of them than to those who do not: “For unto every one that hath shall be given, and he shall have abundance: but from him that hath not shall be taken away even that which he hath.” Since the performance of fine-tuned LLMs is positively correlated with the volume of data available to train them, the Matthew effect means that proficient and prolific writers, who already enjoy the benefits of an established publication record, can benefit disproportionately from personalized LLMs. This is clearly demonstrated in the difference in performance between our models trained on few (SPM) versus many (BDE, JS, CO) previous publications (see [Supplementary Information](#)). The extent to which this can be addressed by creating synthetic text based on existing training data or other means remain open questions.

Another ethical concern is related to the potential homogenizing effect of LLMs on writing styles. At the individual level, reliance on personalized LLMs might prevent or inhibit the further evolution of writing styles. At the collective level, models adapting newly generated or formatted outputs to a specific style may inadvertently reduce diversity and potentially diminish the authenticity of LLM-produced texts as well as the enjoyment gained from reading them (Dwivedi et al. 2023).

Issues surrounding consent and the use of publications as training data are particularly salient for fine-

tuned models. While the use of writings of which one is the sole author seems to us to be ethically unproblematic, this is not necessarily the case for coauthored publications (see discussion on this point below). Even more suspect is the use of writings to which one has not contributed at all. Using an individual's writing style or data without permission might violate privacy and intellectual property rights. However, issues of authenticity and the assignment of credit might remain contentious even in cases where consent has been obtained, but those using the resulting fine-tuned models have not themselves contributed to its training data.

Relatedly, misusing LLMs for various purposes, particularly style theft, poses significant ethical risks. Given the significant market opportunity for assisting individuals or companies with fine tuning models, websites such as chatbase.co and reword.com are already making it possible to easily personalize LLMs based on someone's writings at cheap or no cost.⁹ A malicious actor could therefore easily train and exploit personalized LLMs without consent for plagiarism or fraud, compromising academic integrity or professional reputation.

Given that AUTOGEN appears to perform better on topics similar to those on which it was trained, there is a risk that access to AUTOGEN-type models would perversely incentivize scholars to focus on work that is only marginally different to what they have written before, while still crossing some minimal threshold of publishability. This may be analogous to so-called "salami slicing"—the tendency to take a set of studies or ideas and slice them into as many tiny, incrementally advancing papers as one can, just to get more publications, when really, they should all be in the same paper. Similarly, the higher performance on familiar topics of AUTOGEN might incentivize scholars to produce ever more papers, each one only marginally different from the last, rather than to make bigger, more substantive leaps of scholarly or intellectual progress. Although this is likely to be a genuine problem, especially given the already unmanageable volume of papers published in many disciplines each year, there may be practical ways around this. For example, it might become an academic norm or even a policy requirement that one can use AUTOGEN to generate introductions, conclusions, or background sections in which a more or less rote synthesis of

existing ideas and scholarship is necessary, while still needing to manually develop the bulk of the substantively new material.

OPEN ETHICAL QUESTIONS

LLM use by humans, like human use of generative AI more generally, entails a moral credit-blame asymmetry (Porsdam Mann et al. 2023). On the one hand, blameworthiness for harmful content, or other bad outcomes caused by human use of LLMs, arguably only requires recklessness or negligence (a low bar). On the other hand, and by contrast, creditworthiness for good or valuable outcomes in such a context requires that one invest substantial effort, talent, or sacrifice—specifically of a kind that may be directly undermined or diminished by the use of LLMs (a high bar). Thus, individuals may be blameworthy for low-effort, unskilled, yet careless or reckless use of LLMs to generate bad or harmful content, but cannot similarly take credit for good or valuable content generated through a similar (low) investment of effort or skill.

Personalized LLMs like AUTOGEN complicate this issue. Insofar as such models facilitate the production of valuable writing, their use may diminish the effort and skill put into the writing process and thus, arguably, creditworthiness. However, unlike more general, non-personalized models, personalized models trained on one's own prior writing represent the outcome of previous applications of labor, skill and sacrifice. Perhaps, then, personalized LLM use occupies a middle ground with respect to praiseworthiness: between the generation of text by non-personalized models (least credit) to entirely human-generated writing of text (most credit). Insofar as we care about human moral creditworthiness, that may favor training and using personalized, over non-personalized, models for text generation.

This position seems plausible to us in the case of LLMs trained by an individual using only their own past writings. However, when the LLM is trained on a mixed dataset with coauthored texts or even other authors' works, more complicated questions of ownership and attribution arise. One set of issues concern the proportion of text written by a specific author of a multi-author paper. The approach we took in training AUTOGEN was to use as training data only those publications on which SPM, BDE, or JS are first authors, as in the bioethical context first authorship is typically a reasonable proxy for main drafting responsibilities. This assumption may not hold for other fields, such as law, where coauthor rankings may be

⁹It should be noted that these services offer a different type of personalization than that described here. So far as we are aware, there are not yet any services offering fine-tuning on previous academic output in the way we have here described.

determined by seniority or alphabetical order. As the proportion of published text attributable to the author seeking its use for LLM customization decreases, the moral importance of obtaining the consent of coauthors increases.

In cases where an individual is using a model not owned by them, the output's ownership becomes even more challenging to determine. Licensing agreements can play a critical role in clarifying rights and responsibilities associated with using LLMs, specifying who should be credited and under what circumstances. However, the extent to which certain licensing arrangements should be considered legitimate for authorship ascription remains an open question that should be discussed as early as possible. For example, roughly half of the training data used in the combined AUTOGEN model derives from JS's writings; the large majority of the rest derives from BDE's writings; and only a small minority from SPM's writings. Should SPM be allowed to license AUTOGEN-CO? If not, what percentage of contribution should serve as a cut-off value? If yes, what about someone who has contributed nothing to the training data?

Further pressing questions concern whether data contributors should be listed as coauthor. This might depend on the level of their contribution to the model, and their role in the creation of the output. If someone's work significantly impacts the final output, coauthorship might be justified. However, listing someone as a coauthor—or even as text contributor—who has not personally written any material in the text seems problematic. Doing so without their consent could lead to further ethical concerns.

Additionally, the risk of plagiarizing arises when the LLM generates content that closely resembles the original works used in training. Ensuring that the output is sufficiently distinct from the source material is crucial to maintaining ethical standards in academic writing.

OPEN TECHNICAL QUESTIONS

There are also numerous open questions relating to more technical matters. In presenting our findings, we have only shown unedited AUTOGEN completions using the fixed input format described in the methods section above. However, other methods of using AUTOGEN are possible and we are exploring these in ongoing work. For example, completions can be made to continue manually written text by including leading sentences in the prompt. By writing out the first or more sentences in a given section, and including these in the prompt, AUTOGEN can be made to continue

the line of thought expressed in those sentences. Similarly, AUTOGEN outputs can be edited and fed back into the prompt. For example, if AUTOGEN provides a five-paragraph completion but only the first paragraph is on-topic or otherwise interesting, the last four paragraphs can be deleted and AUTOGEN can be made to generate new completions continuing from the first paragraph.

Furthermore, and very importantly, AUTOGEN can be used as a discrete step in a wider workflow. As mentioned, AUTOGEN is based on the GPT-3 model, which shows much inferior apparent reasoning abilities than the successor GPT-4 model (Bubeck et al. 2023). However, AUTOGEN and GPT-4 can be easily combined. The rough material generated by AUTOGEN could be fed into GPT-4 for improvement. For example, we have had promising early results using prompts such as "Fix the format of the following text and correct errors of reasoning and structure. Based on the following text and abstract, write an introduction to a scholarly paper on [x]." GPT-4 can also be used to rewrite AUTOGEN completions according to specific instructions, such as "Rewrite the above. Get rid of the second objection and combine the third and fourth into one."

The potential benefits of integrating both AUTOGEN and GPT-4 into a workflow have recently been expanded by the release of plugins and browsing for GPT-4, and the ongoing release of the Code Interpreter. This allows GPT-4 to access and process information from the internet (browsing), third-party applications ("plug-ins"), and manually uploaded documents in a variety of formats (Code Interpreter). Other developments, such as AutoGPT, expand these capabilities even further by allowing individuals to assign multiple goals and tasks to LLMs which then attempt to carry these out without further instruction. Thus, the completions we present above could likely be substantially and easily improved using other LLMs such as GPT-4.

CONCLUSION

We introduce the use of a fine-tuned, personalized LLM for academic writing, which we dubbed AUTOGEN. Though our experiments are preliminary and informal, we believe they are sufficient to demonstrate the potential benefits of LLM fine-tuning on scholars' own prior publications, especially for the generation of text and relevant ideas close to the subject matter of included writings.

The potential benefits, such as increasing academic efficiency and preserving writing styles, must be weighed against serious concerns including intellectual property rights, privacy, and the potential for misuse. Attempts to regulate personalized LLMs at both societal and institutional levels should address how to mitigate these concerns whilst pursuing benefits associated with the opportunities. Further, the ethical complexities surrounding authorship and credit attribution, particularly for multi-source trained models, present novel challenges that remain to be worked out.

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